

Climate Change Policy Preferences and Foreign State Behavior – Survey Experimental Evidence on Reciprocal Defection from Switzerland *

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Abstract

For a long time, scholars and public debate alike identified the structure of the climate crisis as a collective action problem as the predominant challenge to effectively reduce greenhouse gas emissions worldwide. Recent scholarly work, however, proposes that existing evidence is not consistent with collective action being the binding constraint (Aklin and Mildenberger 2020) – a claim that Kennard and Schnakenberg (2023) revisit and challenge. We bring new evidence to this question, drawing on data from a high-quality population representative survey experiment with the Swiss citizenry ($N = 3502$), focusing on whether and to what extent citizens react to signals of large foreign countries defecting from climate policy as agreed under the Paris agenda. Our survey-experimental approach allows us to work with counterfactual scenarios to identify this effect. We show that foreign state behavior matters for preferred domestic climate policy ambition levels; in particular, defection is reciprocated strongly, even among subgroups likely to show a potential counterbalancing logic. Importantly, ego- or sociotropic concerns for economic losses with the green transformation strongly increase this taste for reciprocal defection. Altogether, this indicates that the perspective of collective action theory should not be discarded as a relevant constraint for effective climate policy.

Keywords: collective action theory; free-riding; reciprocity; preferences; public opinion; climate policy

Wordcount: 5,498

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1 Introduction

Countries around the world are failing to address the climate problem – despite sufficient knowledge, long-term economic and environmental benefits of taking action, and technological prerequisites available to take action (Calvin et al. 2023). Why is this the case? For a long time, scholars and public debate alike identified the structure of the climate crisis as a collective action problem as the predominant challenge to effectively reduce greenhouse gas emissions worldwide (e.g., Bernauer 2013). Given that the costs of any additional ton of greenhouse gas emissions are externalized to all countries (and their citizens) around the globe, state-level incentives to sincerely engage mitigation policy seem insufficient. However, this collective action problem does not only characterize the international, but also the national level, (Kennard and Schnakenberg (2023), henceforth KS): given that curbing emissions requires costly behavioral change, any substate actor has incentives to invest in passing on any potential burden – implying distributive conflict at the national level as a second relevant constraint to consider. Aklin and Mildemberger (2020) (henceforth AM) discuss this dual problem structure in detail. They propose that existing evidence is not consistent with collective action being the binding constraint – a claim that Kennard and Schnakenberg (2023) revisit and challenge (see also Aklin and Mildemberger 2023). These contributions agree, however, that it is of utmost necessity to bring additional evidence to this question – in order to focus scarce political and scholarly resources onto lifting relevant constraints.

In this article, we contribute survey-experimental evidence from Switzerland to understanding the relevance of free-riding concerns. Importantly, we rely on data from a high-quality on- and offline panel of the Swiss resident population > 16 years (the ‘Swiss Environmental Panel’), allowing for direct generalization to the Swiss citizenry. Substantively, we investigate a major focus of theoretical literature on the public goods

structure of the climate problem at the international level: reactions to the defection of major players, and whether they reciprocate defection with defection (as implied, e.g., by a grim trigger strategy). Investigating this matter with survey experiments is a fruitful research strategy: as KS highlighted, solutions to the repeated prisoners' dilemma arise theoretically because of actors' consideration of counterfactual scenarios, which are however observationally hidden – methodically, survey experiments can directly work with such counterfactual scenarios, however.

Inferring the relevance of free-riding concerns at the international levels (where governments act) based on citizen preferences of course relies on the crucial assumption of a close link between citizen preferences and government behavior – which seems plausible given recent evidence (see summary in Rudolph, Freitag and Thurner (2023: A.4)): public preferences matter for foreign policy-making (Goldsmith and Horiuchi 2012), and studies from developed economies indicate that decision-makers take the environmental preferences of voters into account when deciding on environmental policy (e.g., Vandeweerd, Kerremans and Cohn 2016; Anderson, Böhmelt and Ward 2017; Weaver 2008). More generally, research indicates that public opinion can shape policy debates, the extent of political mandates, and subsequently the decision-making of policymakers (Burstein 2003; Page and Shapiro 1983; Wlezien 1995). For example, Hager and Hilbig (2020) show that exposure of politicians to polling information affects both their priorities and positions in political speech. Hence, policy seems to reflect shifts in public sentiment (Bakaki, Böhmelt and Ward 2020; Schaffer, Oehl and Bernauer 2022), and it makes sense to infer the relevance of state-level constraints from public opinion survey experiments.

2 Argument

We start the theoretical argument by outlining some descriptive evidence on climate preferences and free-riding concerns among the Swiss population: in 2019, respondents are overall rather skeptical whether Switzerland will, and also whether it even can reach its emission reduction goals.¹ However, when citizens subsequently rate the likelihood of four potential international consequences of Switzerland failing to reach its emission reduction targets, respondents perceive reciprocal defection by other countries both as relatively and absolutely likely to occur, with a mean of 3.41 (median of 4) on the 5-point scale (see Figure 1). In turn, they assess other potential consequences that were inquired, lowered reputation (2.81, median of 3), less willingness to cooperate with Switzerland (2.38, median of 2) or even trade sanctions against Switzerland (2.21, median of 2) as rather unlikely to occur. Hence, when judging the consequences of their own country's behavior, the idea of a dilemma-type situation with reciprocal defection (though not punishment) resonates most with citizens. But would citizens apply the very same logic to preferred reactions to other countries' behavior?

¹ Mean of 3.36 that these will be reached; mean of 3.11 they can be reached; scale 1 (certainly) to 5 (certainly not). See Appendix Figure A.1 for the respective distributions.

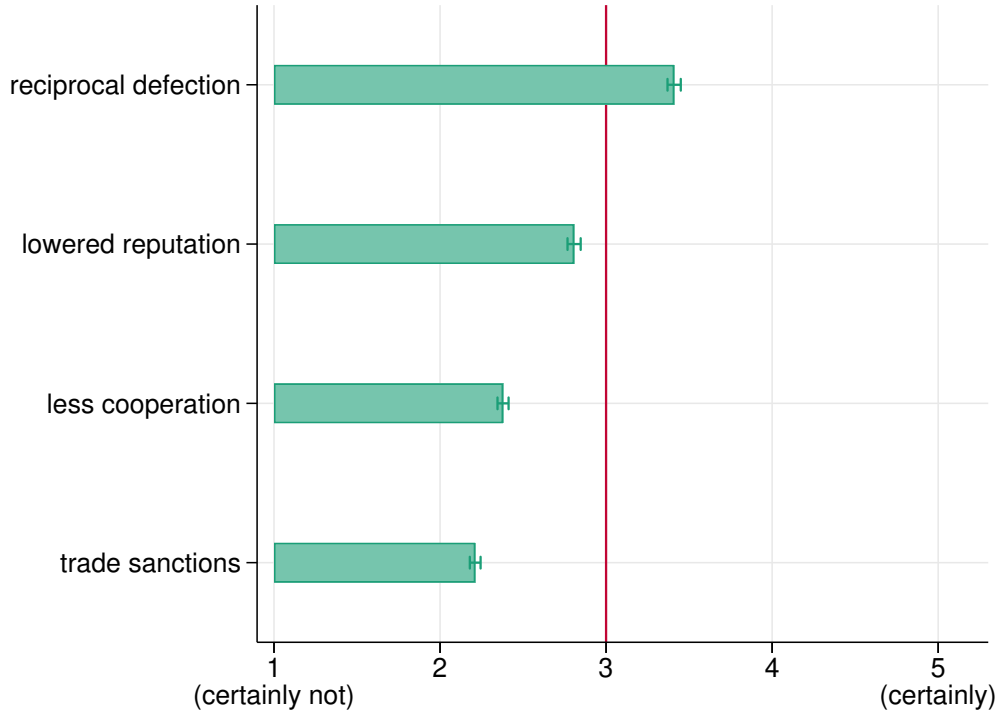


Figure 1: Average rating on “What international consequences would you expect or not expect if Switzerland did not achieve its goal of reducing its greenhouse gas emissions by 50% by 2030 and at least 70% by 2050?” with a scale from 1 (“Certainly not occurring”) to 5 (“certainly occurring”) for four items (a: “Switzerland’s reputation in the world would be damaged.” B: “Other countries would restrict their trade with Switzerland (trade sanctions).” c: “Other countries would cooperate less with Switzerland in international bodies.” d: “Other countries would not reduce their greenhouse gas emissions either.” ; $N = 3,145$

This is both theoretically and empirically ambiguous on two fronts. First, do citizens apply a reciprocal logic at all, i.e., do citizens respond to a reciprocal logic at the international level in climate policy support? Theoretically, others have shown that there is good reason to expect that citizens respond to foreign state behavior (see detailed summary in AM and KS). We take from their discussion the general expectation that

Hypothesis 1 *The probability of citizens supporting costly climate action will be affected*

*by the behavior of other countries with respect to climate policy.*²

Notably, AM (p.15f.) summarize evidence for this expectation (relevant studies they draw on are, e.g., Tingley and Tomz (2014); Bechtel and Scheve (2013); Li et al. (2004); Tvinnereim, Lachapelle and Borick (2016)), and conclude that we overall see no relevant evidence in this direction: “an emerging climate opinion literature has found scant evidence that national publics are conditional cooperators” (AM:14). Their own assessment, going beyond the citizen perspective, adds to this picture. They propose that overall climate policy of states is not affected by foreign state behavior (but rather by domestic distributional concerns). However, this view is contested. In particular, KS (p.6) propose this stems from “a highly selective reading of the literature [and that] a number of important studies provide empirical evidence which directly contradicts the conclusions of Aklin and Mildenberger (2020)” – referencing Bechtel, Scheve and van Lieshout (2019); Bechtel and Scheve (2013); Tingley and Tomz (2014). Indeed, a search for literature since 2020 also indicates that while some work dismisses the relevance of free-riding concerns for preferences on mitigation policy (e.g., Gazmararian and Milner (2022)³), others show strong evidence that other countries’ behavior is taken into account by citizens (e.g., Bechtel, Scheve and van Lieshout (2022: 1)⁴) – altogether, there is surprisingly little recent research on the direct relevance of other state’s behavior for domestic preferences, despite this important scholarly debate. This calls for additional evidence on the matter whether and how citizens react to foreign state behavior at all.

Second, if citizens respond, the question is whether defection [increased cooperation]

² Note that this hypothesis is more general than the directed hypotheses (their WCA2 and WCA3) of AM.

³ They propose that not free-riding, but projected affectedness by adverse climate outcomes drives ambition levels.

⁴ They show that other countries’ engagement increases effectiveness beliefs and the perceived fairness of pro-climate policy, such that “multilateralism significantly increases public approval of costly climate action”.

by other states leads to support for reciprocal defection [reciprocal cooperation] in a tit-for-tat logic – or whether citizens respond by support for counterbalancing strategies. Overall, the literature usually investigates a mutually reinforcing logic, i.e., that support for ambitious climate reduction policy increases if citizens learn that other countries ramp up their efforts (and vice versa). Notably, however, some results of Tingley and Tomz (2014) indicate that at least for a subset of the population conditional cooperation might also be defined the other way around: supplementing lack of climate policy of other countries if they learn these countries are defecting. This reflects in both AM and KS, while both starting from general public goods theory, coming to vastly differing theoretical conclusions what to expect here in the first place. In particular, AM proposes that other countries engaging [defecting from] climate change mitigation policy *increases* [*decreases*] a countries probability of doing the same (see their prediction WCA2 and WCA3). In turn, based on a formal model, KS (p.11) propose the opposite relationship between the right- and left-hand side of this relationship, as “climate actions are strategic substitutes rather than strategic complements.” As Aklin and Mildemberger (2023: 148) emphasize, this latter model and its predictions “depart from conventional models of global climate politics that emphasize reciprocal defection motivated by free-riding concerns.” Hence, this is both theoretically and empirically worthwhile to investigate. We want to directly test this relationship focusing on the *consequences of defection of pivotal players*. Subsequent the model of KS we expect that

Hypothesis 2a *Pivotal players defecting from climate change mitigation increases support for climate change mitigation policy.*

Subsequent the reading of AM we should expect that

Hypothesis 2b *Pivotal players defecting from climate change mitigation decreases support for climate change mitigation policy.*

By drawing on a survey experimental approach, we can directly engage the necessary counterfactual we need to investigate this relationship – besides observational evidence of Tingley and Tomz (2014) and anecdotal evidence (see summary in Aklin and Mildenberger (2023)) we are not aware of a rigorous test of this prediction.

Last but not least, besides contributing evidence for this important theoretical debate, we investigate whether potential heterogeneity in citizen’s motivations to react to other countries’ behavior. In particular, we propose that reciprocity could be fueled not only “because of reduced free-riding concerns”, but also be moderated by expectations on domestic benefits from embracing vs. delaying a carbon-neutral economy (see also Bayer and Genovese (2020)). Such domestic considerations provide for a new mechanism by which constraints at the international and the domestic level could be connected. Of course, egotropic considerations have long been shown to shape policy preferences of citizens (e.g., Anderson (2007)), while recent research has started to also investigate to what extent sociotropic considerations matter (e.g., Schaffer and Spilker (2019)). Particularly, we would expect that

Hypothesis 3a *Citizens that perceive themselves as economically disadvantaged by climate change mitigation policy are more likely to prefer to reciprocate defection by large polluters with defection.*

Hypothesis 3b *Citizens that perceive their country to be economically disadvantaged by climate change mitigation policy are more likely to prefer to reciprocate defection by large polluters with defection.*

3 Research Design

Our data comes from the Swiss Environmental Panel (SEP), a large on- and offline panel survey, carried out since spring 2018 (Rudolph et al. 2020). The sampling is conducted by the Swiss Federal Statistical Office (BfS), and drawn to be representative of the Swiss resident population aged 15+. Wave 2 of this panel, on which the present analysis focuses, deals with climate change preferences, fielded in winter 2018/2019, with a median survey duration of 13 minutes. The sample size of the second wave is 3,502, 73% of wave 1 respondents (first wave: 4,813, raw response rate of 32.17%).⁵ As the sample of invitees is a random draw of the Swiss resident population, and hence population-representative, and as Rudolph, Quöß and Bernauer (2020) document that survey take-up shows no strong correlation between register-verified and census-verified socio-demographic characteristics, we are confident that we report results that our results speak for the general Swiss population.

3.1 Set up of the survey experiment and core dependent variable

Our core dependent variable is an ordinal choice on a policy preference scale on whether “Concerning its own planned measures to reduce greenhouse gas emissions by 50% by 2030 and at least 70% by 2050, Switzerland should...” “Further enhance them” (level 1), “Continue to fully implement them” (level 2), “Implement them in a weakened form” (level 3), “Stop them, but continue to participate in the Paris Climate Accord” (level 4), “Stop them and withdraw from the Paris Climate Accord” (level 5). Respondents could also voice “Don’t know/no opinion” (coded as missing in all regression analyses).

⁵ Further information on the panel and access to data is available at doi:10.23662/FORS-DS-1220-1.

We bring choice on this variable together with a randomly assigned binary textual vignette (Mutz 2011), i.e., a split-sample survey experiment, communicating the hypothetical compliance behavior of other countries. In particular, we introduced respondents to a scenario where large polluters comply, or not comply with their measures as announced under the Paris Agreement. This was worded as: “If it turns out in a few years’ time that large countries such as China, India and the USA will {[achieve] // [not achieve]} their announced greenhouse gas reduction targets, what do you think Switzerland should do?”. We code this experimental stimulus as a binary variable “compliance” (0) vs. “defection” (1).

We test Hypothesis 1 by investigating whether vignette conditions affect the mean or distribution of answers to our core dependent variable at all. We test Hypotheses 2a and 2b by running ordinal logistic regressions to test whether being exposed to the scenario of “defection” increases respondents probability to agree with full implementation or even enhanced efforts (H2a), or whether “defection” decreases this probability (H2b).

3.2 Ego- and sociotropic concerns as moderators

To investigate whether domestic considerations moderate the relevance of international behavior, we measure egotropic and sociotropic concerns related to implementing the Paris Agreement.

Concerning egotropic concerns, we assess whether respondents “[...] think that measures to reduce Swiss greenhouse gas emissions by 50% until 2030 will cause higher, equal or lower costs for your household than for the average Swiss household?”, with a three-categorical response scale (“Higher costs” (1), “Same costs” (2), or “Lower costs” (3)). Respondents could also voice “Don’t know/no opinion” (coded as missing in all regression analyses).

Concerning sociotropic concerns, we propose that we need to differentiate two scenarios, sociotropic considerations in case Switzerland *does* comply with the Paris agreement compared to when it does *not*. This mirrors the tension in the public debate with on the one hand supporters of a green transformation regularly emphasizing economic benefits from a switch to greenhouse gas efficient production and consumption (e.g., innovation benefits; industry laggard effects) (Dimitrov 2012); while on the other hand opponents of a green transformation emphasizing the economic burden of a green transformation on the existing business model Drews and Van Den Bergh (2016). We hence inquire both dimensions by way of a split-survey experiment: “In your opinion, will it benefit or harm the Swiss economy as a whole if Switzerland [implements] // [does not implement] its target under the Paris Climate Convention (50% reduction in emissions by 2030, at least 70% by 2050)?”, with responses on a 5-point likert scale from 1 (“Strongly benefit”) to 5 (“Strongly harm”). Respondents could also voice “Don’t know/no opinion” (coded as missing in all regression analyses). This allows us to analyze reactions to “defection” by two types of sociotropic concerns: first, among respondents that belief that implementing the Paris Agreement actually benefits (vs. harms) the Swiss economy; second, which does not necessarily mirror the above, whether defecting from the Paris Agreement actually benefits (vs. harms) the Swiss economy. Given we used a split-sample design for this question also, how respondents answered in one of these scenarios can not influence replies to the other.⁶

Regarding Hypothesis 3a, we expect that respondents that perceive relatively higher costs for their household from Switzerland implementing the Paris Agreement show stronger indications for reciprocal defection compared to individuals that perceive average or even lower costs. We test this by regressing our core dependent variable on the

⁶ Note that in the flow of the survey, sociotropic assessments were asked before the international cooperation question and egotropic assessments.

treatment indicator, cost perceptions, and their interaction.

Regarding Hypothesis 3b, we expect that respondents that expect harm for the Swiss economy from Switzerland implementing the Paris Agreement should show stronger preferences for reciprocal defection compared to individuals that expect benefits – likewise, respondents that expect benefits for the Swiss economy from Switzerland *not* implementing the Paris Agreement should show stronger preferences for reciprocal defection compared to individuals that expect harm. We test this by regressing the dependent variable on the treatment indicator, a three-categorical indicator for respective cost expectations ((strong) harm/both harm and benefit/(strong) benefit), and their interaction.

All regression analyses use robust standard errors. Robustness tests report results from linear models.

4 Results

The following section reports our results, turning first to respondent’s reaction to the “defection” vignette, providing evidence for H1 and H2a/2b. Next, we investigate how these are moderated by egotropic (H3a) and sociotropic concerns (H3b) for households/Swiss economic outlook under implementation (or not) of the Paris Agreement.

4.1 Do citizens respond to defection?

As can be seen from Figure 2, irrespective of treatment condition, a large majority the respondents is in favor of implementing or even strengthening Swiss climate policy. However, foreign state compliance (or not) makes a notable difference for the level of ambition that respondents prefer. Under compliance, 54% want to implement as agreed, and 29% even want to strengthen Swiss measures, i.e., an overwhelming majority of 83% supports

Swiss’ Paris targets. This changes markedly if other countries defect: then, it is only 73% of citizens in favor of the Swiss’ Paris targets. The share of “strengthening” drops strongly to only 11% of citizens favoring such policy. While still a sizable share of respondents favors “implementation” (61%), shares of respondents that want to weaken, stop implementing, or even retract from the Paris process altogether increase relevantly (by 5, 3 and 3%-points), leading to opposition to current ambition of 23% under defection as compared to 11% under compliance of large polluters. These results are in line with expectations from Hypothesis 1.

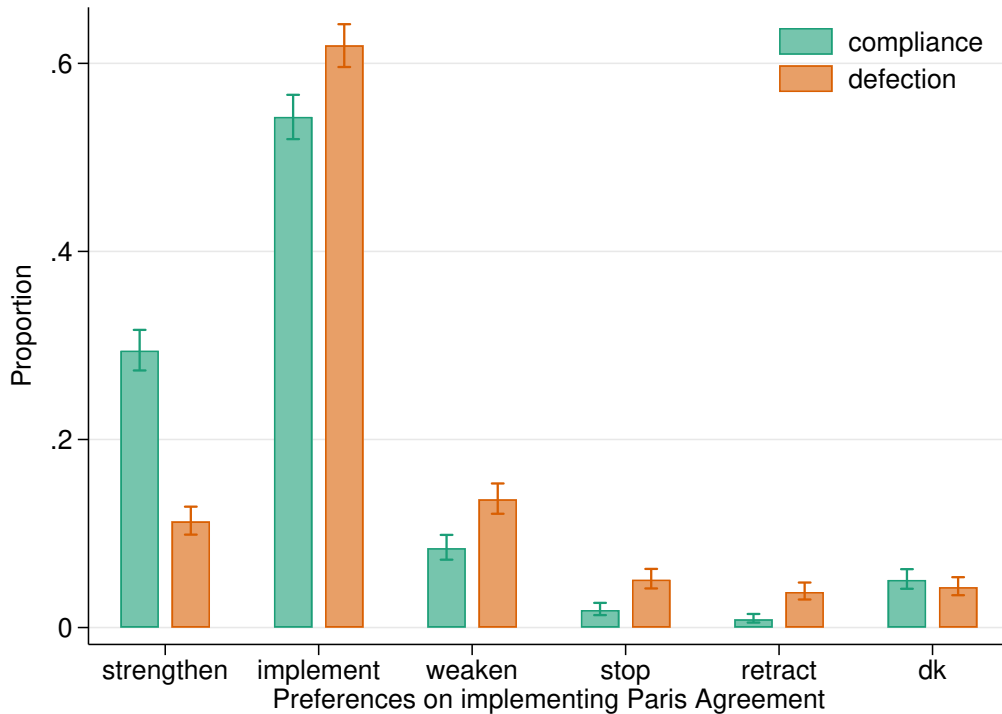


Figure 2: Proportion of respondents supporting either policy option under compliance (green bars) or defection (red bars) of large polluters. Appendix Figures A.2 and A.3 provide corresponding treatment effect coefficients from linear and ordered logit models. $N = 3,464$

Also, these findings indicate that defection by pivotal players strongly reduces citizens’ commitment to ambitious climate change mitigation policy in line with Hypothesis 2b –

both regarding support for overcompliance, and support for reciprocal defection.

We find no support for Hypothesis 2a that defection induces a counterbalancing logic. Given results of Tingley and Tomz (2014) that at least some subgroups of the citizenry could embrace such a counterbalancing logic, we engaged in additional inductive analysis into whether most likely subgroups could show a pattern of reactions in line with H2a. However, as reported in Appendix Figures A.4 and A.5, neither citizens that strongly believe in human-made climate change, nor citizens that support net-zero emissions⁷ show any signs of counterbalancing. While baseline support for the Paris process differs between subgroups, all reciprocate defection by large foreign polluters by way of reducing their support levels for climate mitigation policy (albeit to different degrees). Altogether, we also find no support for Hypothesis 2a within most likely subgroups.

4.2 Evidence for a reinforcing logic?

In a next step, we report our analyses whether a reciprocal defection could be reinforced by concerns for redistributive consequences of compliance (or not) with the Paris agreement. To this end, we differentiate respondents by large vs. small ego- or sociotropic concerns for redistribution from ambitious climate policy. Regarding egotropic concerns, we investigate whether the reaction to the “defect” treatment conditions is stronger among respondents that would feel their household would be particularly burdened by climate policy measures compared to an average Swiss household (vs. not). As reported in Figure 3 the defection treatment has strong effects inducing reduced support for climate policy. According to the model, with defection, the probability of respondents opting for ‘strengthening’ decreases by 16%-points in both ‘burden’ subgroups. However, among those reporting low burden a sizable share shifts to ‘implement as is’ (plus 8%-points),

⁷ As measured by support for the so called direct-democratic proposal “Glacier Initiative”, which called for net-zero emissions by 2050 at the time of the survey.

whereas others increase their support for weakened implementation (5%-points) or even stopping (2%-points) or retracting (1%-points). Among those reporting an above average burden for their household, also the ‘implementing as is’ policy option sees decreased support (by 2%-points), leading to much stronger shifts to ‘weakening’ (by 11%-points), ‘stopping’ (4%-points) and ‘retracting’ (3%-points). Hence, among those fearing to loose out economically from ambitious climate policy, defection by large polluters decreases support for domestic mitigation policy much more strongly – as evident from the net change in support for implementing as is or strengthening (-18 compared to -8%-points) compared to weakening, stopping or retracting (+18 compared to +8%-points).

Regarding sociotropic concerns, we first inquire expected consequences of compliance with emission reductions from respondents. Here, we proposed that we have to differentiate how respondents perceive the implementation or defection of the Paris agreement to relate to the performance of the Swiss economy. Given proponents of implementation emphasize transformational gains, just as opponents highlight transformational losses (Vandeplas et al. 2022; Drews and Van Den Bergh 2016), this approach is warranted. We do this in a survey experimental way, investigating whether respondents expect that implementing or *not* implementing the Paris agreement leads to domestic economic burden, benefit, or both (on a 5-point scale, which we collapsed to three categories).⁸ We can then relate the effect of defection to policy choice within subgroups of expectations on the domestic economy consequences if the Paris Agreement is implemented or not. As reported in Figure 4, expected harm/benefit conditions relate strongly to how the defection treatment affects policy choice. Still, over all subgroups, the defection treatment induces a reduction in support for climate change mitigation policy effort. However, turning first to the left panel of Figure 4, if respondents expect that *not* implementing

⁸ 22% expect benefits, 41% both, 37% harm from *not* implementing; while 38% expect benefits, 43% both and 19% harm from implementing the Paris agreement.

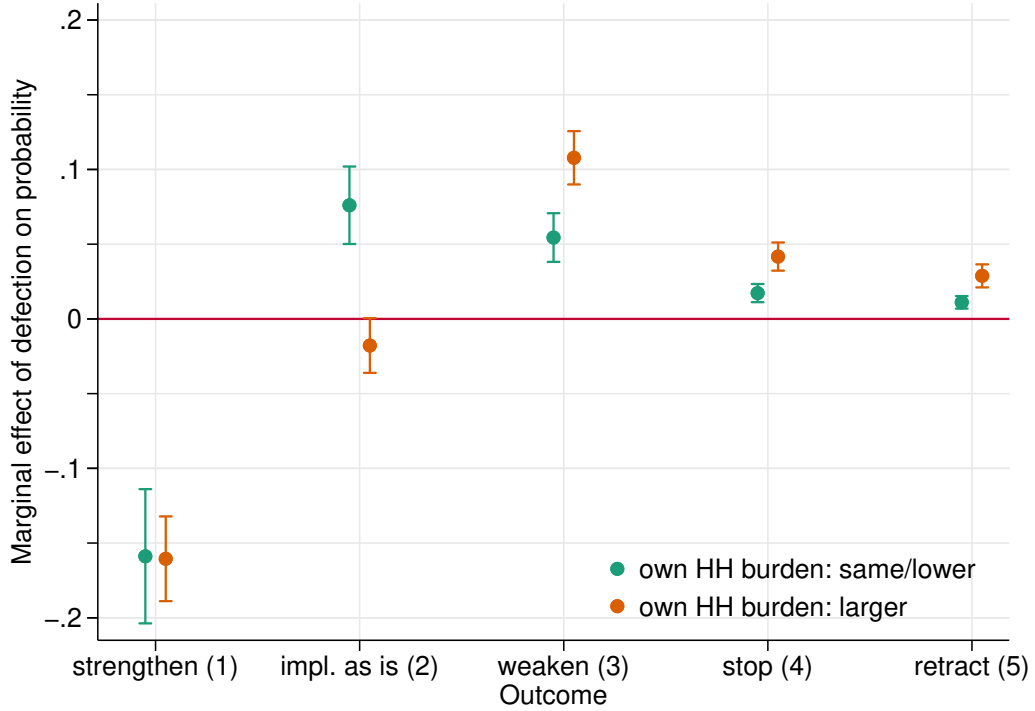


Figure 3: Marginal effects for changes in the probability of choosing one of the outcomes with the survey experimental treatment condition of defection by subgroups of self-reported burden for the respondent's household if climate policy is implemented (1: above average; 0: at or below average). Marginal effects based on ordered logit regression of policy preference on treatment dummy. 95% confidence intervals from robust standard errors shown. $N = 3,124$. Appendix Figure A.6 shows corresponding effects from a linear model.

the Paris agreement harms the domestic economy, learning that large polluters do defect leads to markedly lower support for strengthening domestic efforts (by 30%-points), shifting support to mostly implementation as is (by 14%-points), but also weakening (by 10%-points), stopping (by 4%-points) or retracting (by 3%-points) from the Paris Agreement. This is mainly because of very high baseline support for strengthening climate mitigation policy in this subgroup, at 45% under compliance. Conversely, if respondents expect that *not* implementing the Paris agreement does both harm and benefit, or even benefits the domestic economy, defection reduces support for strengthening efforts by

13%-points, while only marginally affecting support for implementation as is (by 0/4%-points), shifting support to weakening (by 8/10%-points), stopping (by 3/4%-points) or retracting (by 2/3%-points) from the Paris agreement. Taken together, defection by pivotal polluters increases the taste for reciprocal defection overall, and substantially reduces support for increasing climate change policy ambition within a subgroup of citizens that could otherwise carry such policy.

Turning to the right panel of Figure 4, respondents that expect the implementation of Paris agreement policy to harm the domestic economy are strongly reacting to defection. They largely turn to reciprocal defection, with combined support for strengthening or implementing policy as is dropping by 31%-points, and support for weakening (15%-points), stopping (9%-points) or retracting (6%-points) increasing conversely. This pattern also appears for respondents expecting both benefit and harm, though much attenuated (combined support for increasing/implementing as is dropping by only 13%-points) and respondents expecting harm (dropping by 7%-points). Taken together, defection by pivotal polluters again increases the taste for reciprocal defection overall, and substantially reduces support for current (or even stronger) levels of climate change policy ambition within a subgroup of citizens that is potentially skeptical of such policy – at baseline, under compliance, a strong majority of 80% support full implementation/strengthening despite expected harm; this drops to 48% under defection of pivotal polluters. Hence, defection of other countries leads to the emergence of a subgroup showing domestic opposition to Paris-aligned policy on average.

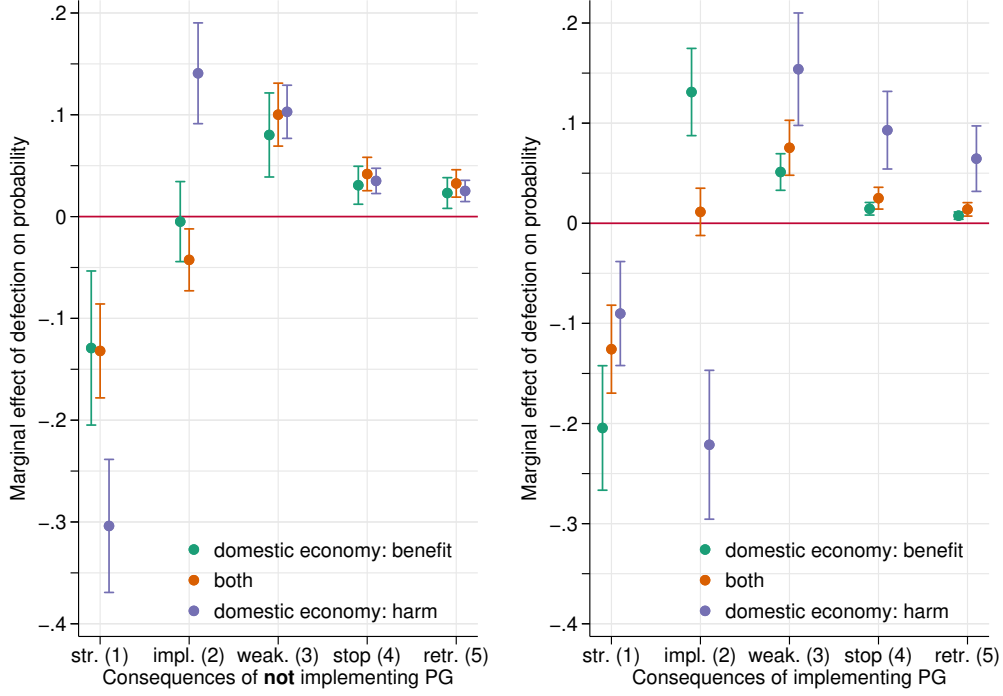


Figure 4: Marginal effects for changes in the probability of choosing one of the outcomes with the survey experimental treatment condition of defection by subgroups of domestic burden, benefit or both (coefficient colors) for the Swiss if climate policy is implemented (right panel) or not implemented (left panel). Marginal effects based on two ordered logit regression of policy preference on treatment dummy, interacted with three-categorical perceived burden extent within subgroups of implementing/not implementing. 95% confidence intervals from robust standard errors shown. $N = 1,698/1,481$
Appendix Figure A.7 shows corresponding effects from a linear model.

5 Discussion

In this article, we report evidence from a survey with Swiss citizens, assessing changes in climate policy preferences when faced with defection as compared to compliance by large polluting states. As Aklin and Mildemberger (2023) summarize, long-standing “debates over global climate institutions have focused [on ...] reciprocal defection after defection by a pivotal player”. This literature lets us expect that citizens responds to such defection,

and particularly that they reciprocate defection with reduced support for ambitious policy at the national level – a view that has been contested empirically by AM (proposing that states, and citizens, do not predominantly react to foreign state defection), and theoretically by KS (proposing that defection could also lead to preferences for compensation). Exploiting a survey-experimental manipulation of foreign state behavior, we show that ‘defection’ is indeed reciprocated, and leads to substantive changes in respondent preferences in our context. Notably, support for policy exceeding Swiss commitments under the Paris agreement drops by 18 percentage points under ‘defection’, and support for falling behind agreed Paris commitments increases by 12 percentage points. This indicates that free-riding concerns matter for citizens’ support for climate policy in line with the Paris targets – and that public good theory should not be discarded as an explanation for why states do or do not implement ambitious climate policy, at least given citizen awareness of foreign state behavior as made explicit in our survey. Even more, we find no support that respondent, not even in most likely subgroups, exhibit a counterbalancing logic, by which they compensate the defection of other countries with increased support for ambitious policy (KS, Tingley and Tomz 2014)

While our survey is not tailored to assess the relative importance of free-riding as compared to distributional concerns, we show evidence that distributional concerns moderate the strength of reactions to defection. Notably, both citizens that expect to personally suffer economically from implementing the Paris Agreement show a stronger taste for reciprocal defection – as do citizens that expect the Swiss economy overall to fare worse with, and better without an ambitious green transformation. As we argue, this is important, as thereby free-riding concerns affect preferences within subgroups particularly important to bring on board when forming pro-climate coalitions – as these are more skeptical to begin with.

Future research could replicate and extend our arguments to other countries and temporal contexts. For example, surrounding the 2019 Swiss national elections, pro-environmental preferences were surging in the electorate (see discussion in Däubler, Quoß and Rudolph (2021)), and as our subgroup analyses show, the strength of baseline preferences matter for the extent of reciprocal defection. On the other hand, preferences of Swiss citizens seem in line with average preferences of citizens in other developed economies both regarding environmental concern (see Rudolph et al. (2022: Appendix A.4.2) and regarding regulatory policy preferences, including pro-environmental provisions (see Rudolph, Kolcava and Bernauer (2023: Appendix A.4)). This lets us expect that free-riding concerns surface in principle to a similar extent in other country contexts and that the perspective of public goods theory should not be discarded as a relevant constraint for effectively mitigating climate change.

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**Online Appendix for
‘Reciprocal defection is a relevant component of
citizens climate change policy preferences, and
moderated by distributional concerns –
survey-experimental evidence from Switzerland’**

Lukas Rudolph¹

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A.1 Descriptive evidence

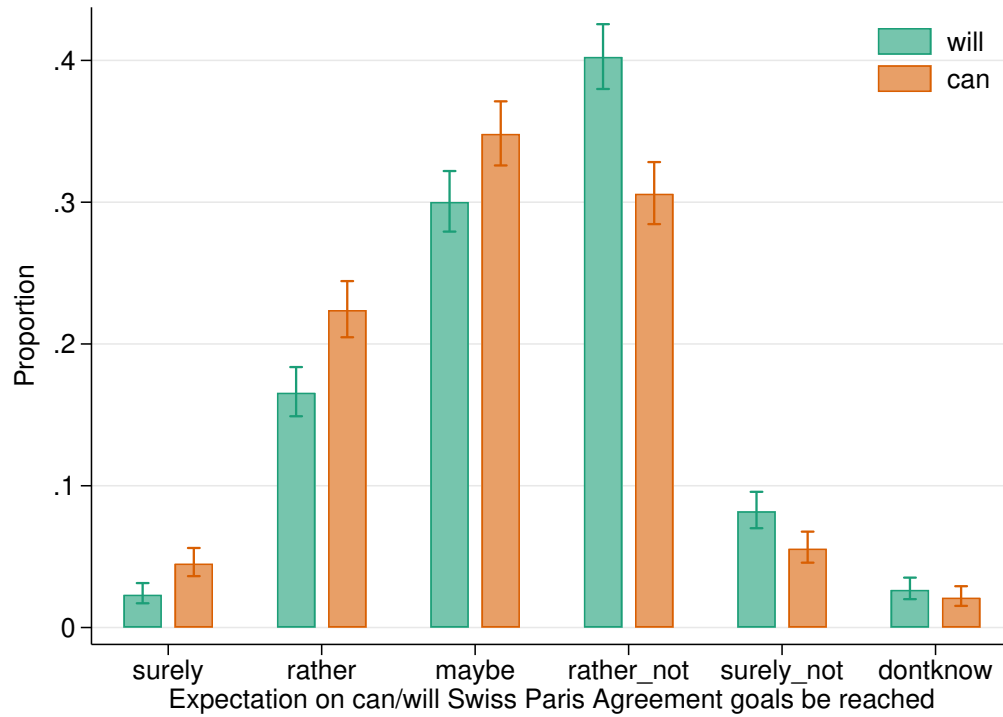


Figure A.1: Distribution of ratings on “Do you think that Switzerland can or cannot achieve the goal of reducing its greenhouse gas emissions by 50% by 2030 and at least 70% by 2050? Switzerland [can reach]//[will reach] this goal...” with a scale from 1 (“Surely reaching”) to 5 (“Surely not reaching”); $N = 1,769/1,706$

A.2 Supporting information for main effect

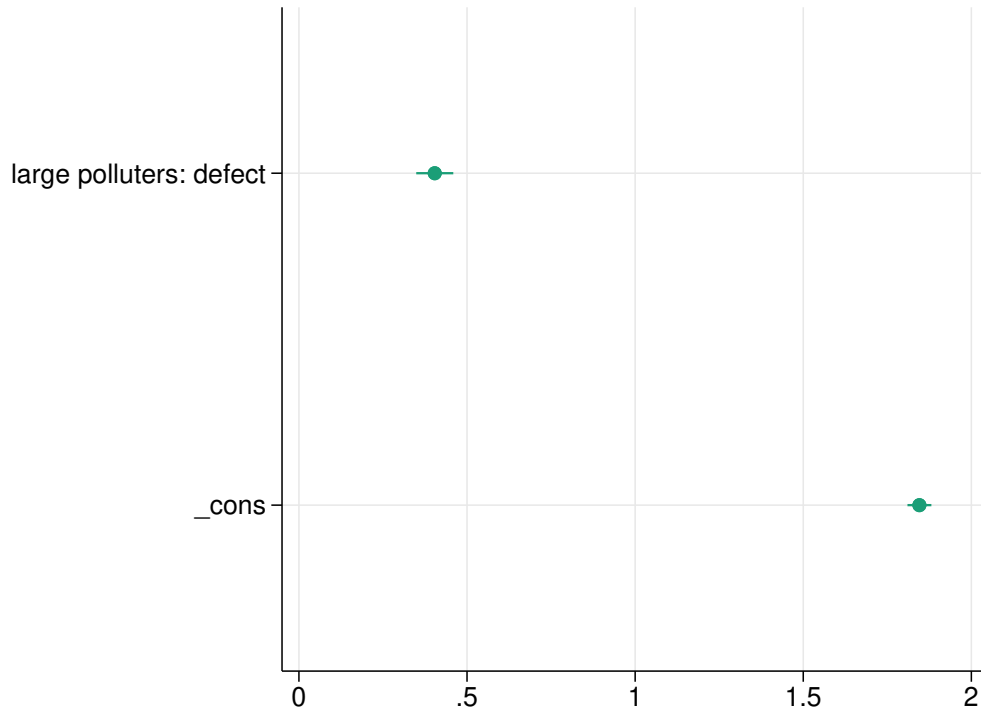


Figure A.2: Constant and treatment effect for survey experimental treatment condition of defection from linear regression of policy preference on treatment dummy. 95% confidence intervals from robust standard errors shown. $N = 3,302$. According to the model, baseline support (under compliance) is at 1.85 on a linear scale (between strengthening and implementing, tilting towards the latter) under compliance, but increases by 0.4 scale points to 2.3 (between implementing and weakening, tilting towards the former) under defection.

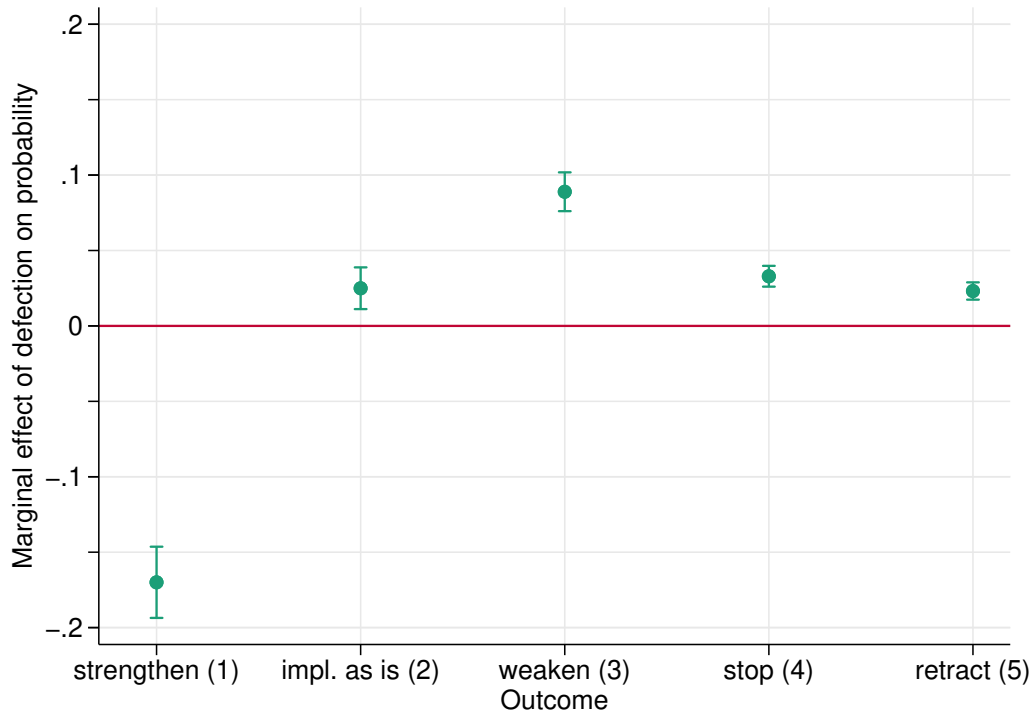


Figure A.3: Marginal effects for changes in the probability of choosing one of the outcomes with the survey experimental treatment condition of defection. Marginal effects based on ordered logit regression of policy preference on treatment dummy. 95% confidence intervals from robust standard errors shown. $N = 3,302$

According to the model, with defection, the probability of respondents opting for ‘strengthening’ decreases by 17 percentage points, while opting for ‘implementing’ as is increases by 2, for ‘weakening’ by 9, for ‘stopping’ by 3 and for ‘retracting’ by 2 percentage points.

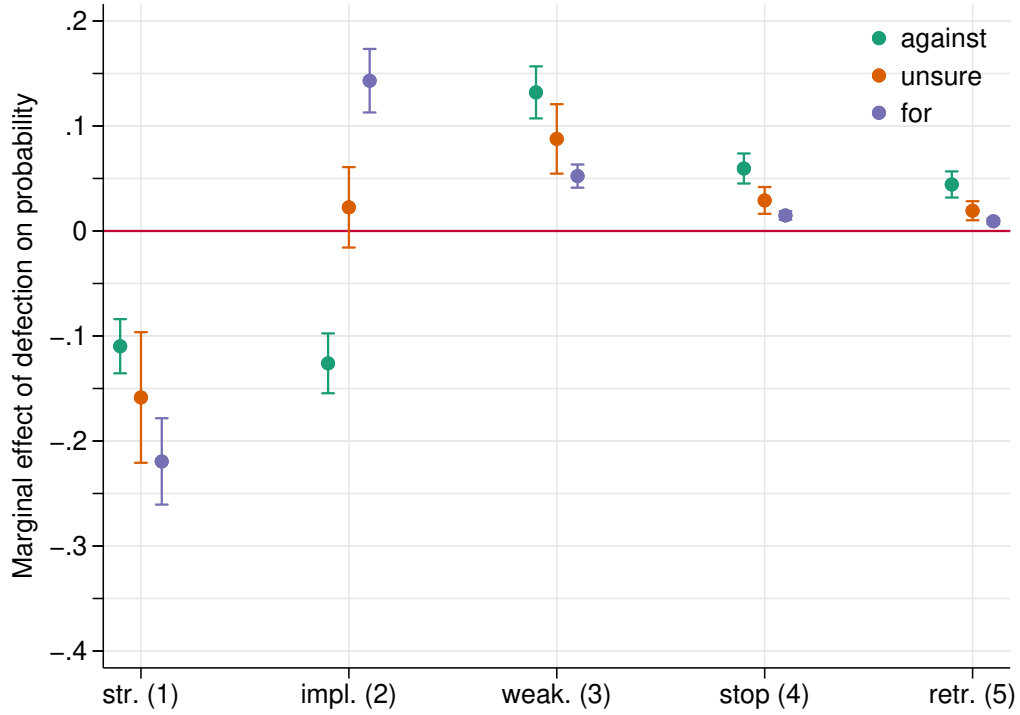


Figure A.4: Marginal effects for changes in the probability of choosing one of the outcomes with the survey experimental treatment condition of defection by subgroups of self-reported support for the ‘Glacier initiative’ (direct democratic proposal demanding net zero by 2050 (2: for; 1: undecided; 0: against)). Marginal effects are based on ordered logit regression of policy preference on treatment dummy, initiative support and interaction. 95% confidence intervals from robust standard errors shown. $N = 3,289$. Results show that support for compliance with the Paris Agreement decreases irrespective of treatment condition, albeit from different baselines, and to different extent.

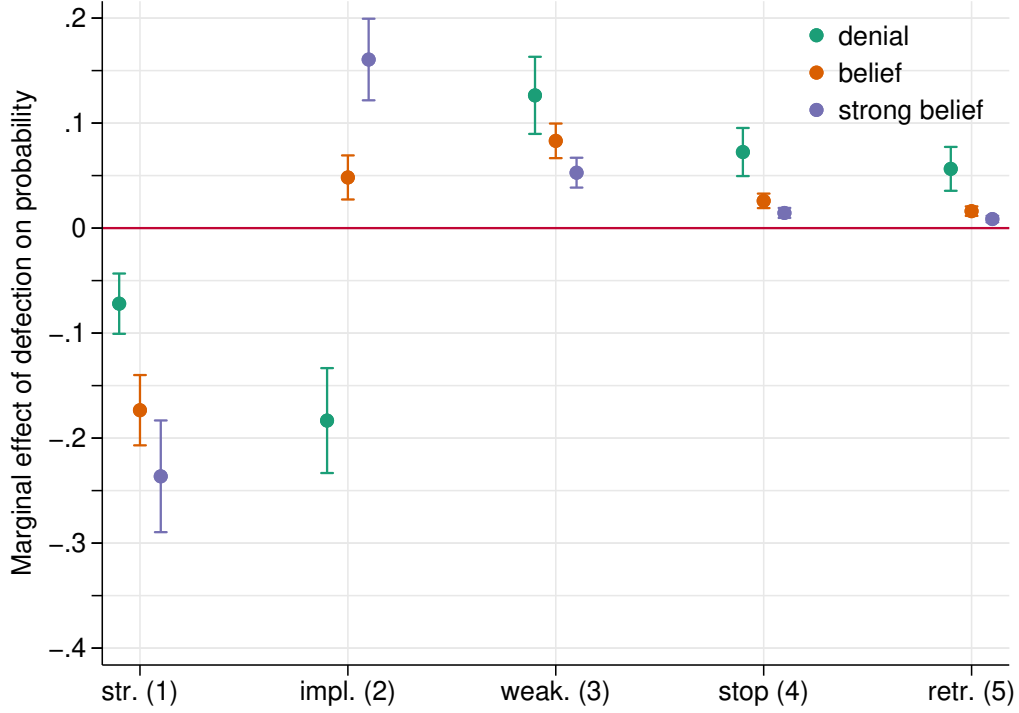


Figure A.5: Marginal effects for changes in the probability of choosing one of the outcomes with the survey experimental treatment condition of defection by subgroups of self-reported climate change denial/belief (denial/belief based on first principle component of replies on four survey items, namely ‘Global warming is a major problem for humanity’, ‘Here in Switzerland, we will be able to adapt to climate change without having to accept a deterioration in our quality of life’, ‘I do not believe that global warming is really happening’, and ‘Global warming is a major problem for humanity’ (all measured on 5-point Likert scales). The bottom 25% of the PCA dimension are labels as exhibiting ‘climate change denial’, the respondents in the 25-75 percentile as ‘belief’ and the top 25% as ‘strong belief’. Marginal effects are based on ordered logit regression of policy preference on treatment dummy, climate change belief category and their interaction. 95% confidence intervals from robust standard errors shown. $N = 3,148$. Results show that support for compliance with the Paris Agreement decreases irrespective of treatment condition, albeit from different baselines, and to different extent.

A.3 Supporting information for Hypothesis 2a

A.4 Supporting information for moderation by egotropic concerns

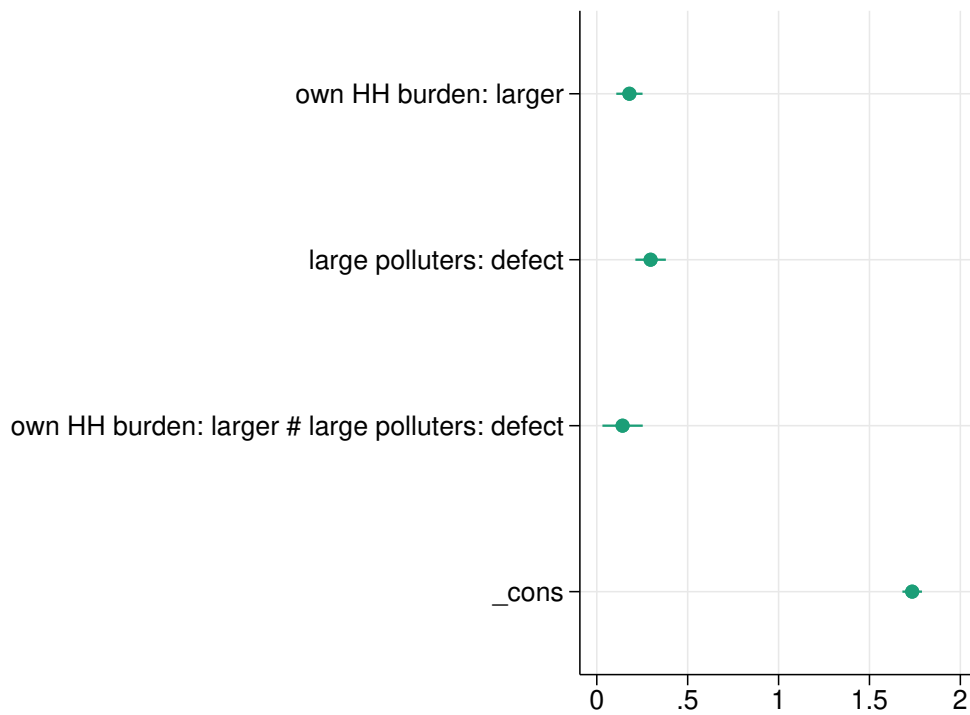


Figure A.6: Constant and coefficients from linear regression of policy preference on household reporting above average burden if climate policy is implemented, defection treatment dummy, and their interaction. 95% confidence intervals from robust standard errors shown. $N = 3,124$.

According to the model, baseline support (under compliance) is at 1.73 on a linear scale under compliance, increases by 0.18 scale points with large burden, by 0.30 scale points with defection, and by another 0.14 scale points with defection given households perceive a large burden.

A.5 Supporting information for moderation by sociotropic concerns

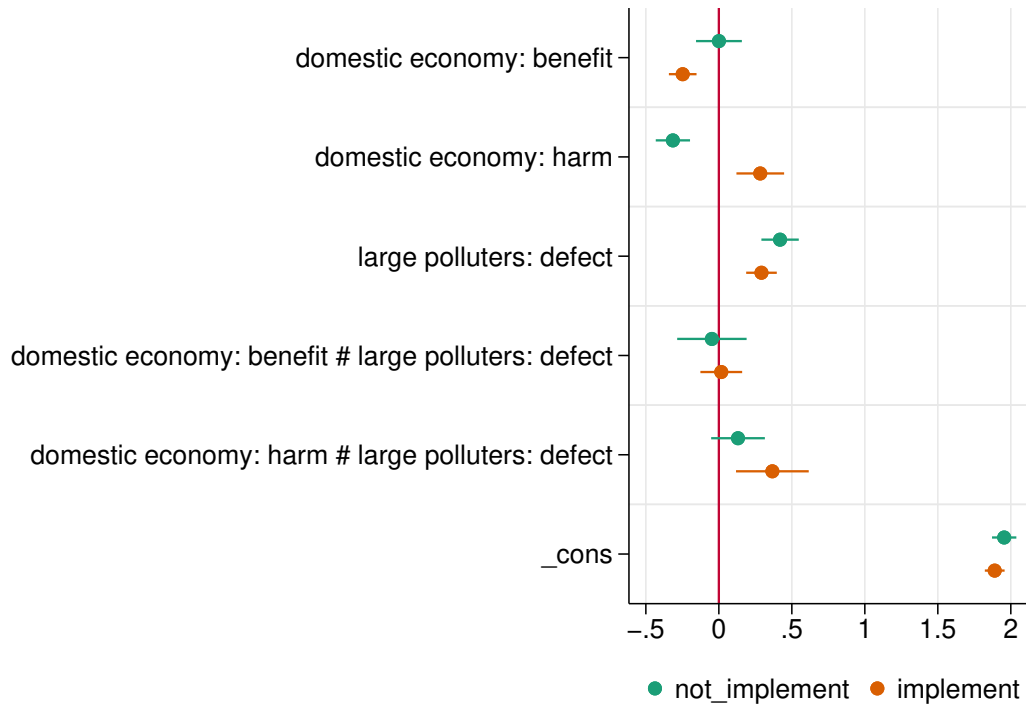


Figure A.7: Constant and coefficients from linear regression of policy preference on household expecting domestic economic benefit/harm (baseline: both) from implementing (model 1) or not implementing (model 2) the Paris agreement, defection treatment dummy, and their interaction. 95% confidence intervals from robust standard errors shown. $N = 1,443/1,629$.